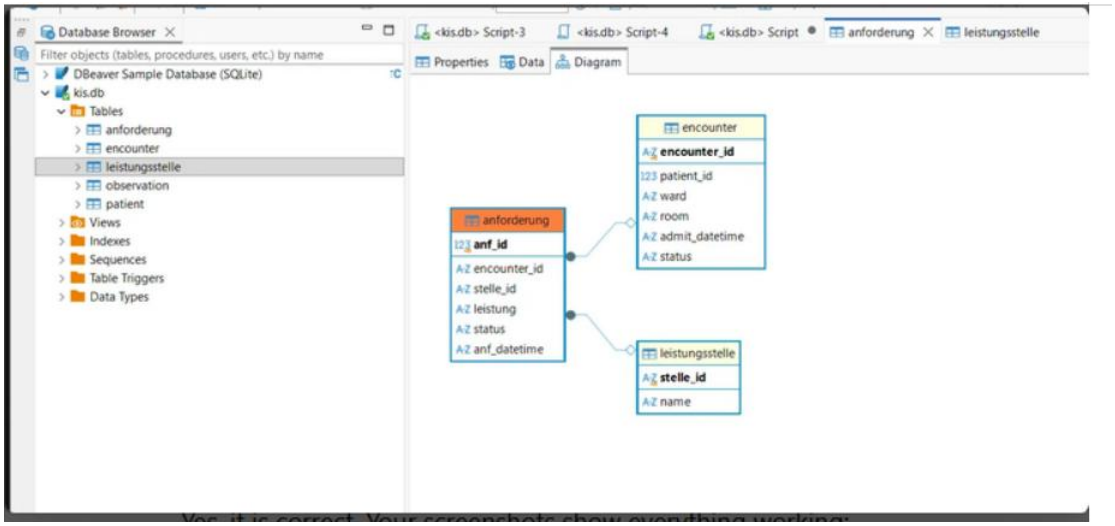
KIS Test Environment: Patient Data Model & LSTM Order Management

DATA MODEL



Workflow

The ward (STA1) orders a blood count (LAB) and a chest X-ray (RAD).

Each order is stored with its encounter and its service unit.

The laboratory completes its order: angefordert $\rightarrow$ abgeschlossen.

The radiology worklist query returns the one open order.

```sql
CREATE TABLE anforderung (
  anf_id       INTEGER PRIMARY KEY AUTOINCREMENT,
  encounter_id TEXT, stelle_id TEXT, leistung TEXT,
  status       TEXT, anf_datetime TEXT,
  FOREIGN KEY (encounter_id) REFERENCES encounter(encounter_id),
  FOREIGN KEY (stelle_id)    REFERENCES leistungsstelle(stelle_id)
);

-- Laboratory completes its order
```

```sql
UPDATE anforderung SET status = 'abgeschlossen' WHERE anf_id = 1;

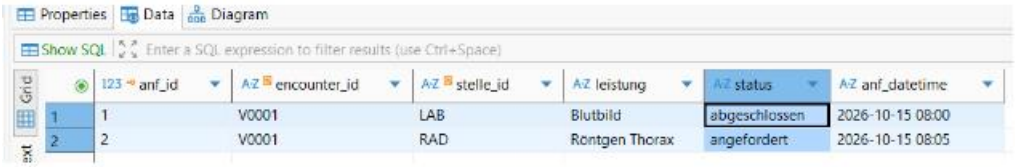
-- Open worklist of radiology
SELECT s.name AS leistungsstelle, p.last_name, e.ward, a.leistung, a.status
FROM anforderung a
JOIN encounter e       ON e.encounter_id = a.encounter_id
JOIN patient p         ON p.patient_id   = e.patient_id
JOIN leistungsstelle s ON s.stelle_id    = a.stelle_id
WHERE s.stelle_id = 'RAD' AND a.status <> 'abgeschlossen';
```

Results All orders (patient, ward, service, service unit, status):

Patient	Ward	Service	Service unit	Status
Mustermann	STA1	Blutbild	Labor	abgeschlossen
Mustermann	STA1	Röntgen Thorax	Radiologie	angefordert

Status per service unit:

Service unit	Orders	Completed	Open
Labor	1	1	0
Radiologie	1	0	**1**



Mapping to HL7 v2 and FHIR R4 (target design)

KIS table	HL7 v2	FHIR R4	Notes
`patient`	PID	Patient	name, birth date, sex
`encounter`	PV1	Encounter	ward, room, admission
`observation`	OBX	Observation	LOINC 8867-4, value, unit
`anforderung`	ORC / OBR	ServiceRequest	`angefordert` → active, `abgeschlossen` → completed

KIS data modelling with referential integrity across patient, encounter, observation and orders.

Order lifecycle logic and service-unit worklists, the principle behind LSTM.

Standards readiness: LOINC coding and a clear path to HL7 v2 and FHIR R4, verified with SQL against expected row counts.

Run it: in DBeaver create a SQLite connection kis.db, open **SQL Editor** $\rightarrow$ **New SQL script**, paste kis_lstm_model.sql

Transparency: this is a self-built test environment with synthetic data. It models the LSTM principle used in hospital information systems and is not a production ORBIS installation.

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